

Hao Li

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RESEARCH INTERESTS

Advancing robotics to achieve human-level learning and adaptability.
Designing bio-inspired tactile sensors for enhanced robotic perception.
Leveraging multisensory integration for robust robot learning and control.
Developing human-centered AI to augment and collaborate with humans.
Creating sim-to-real frameworks for efficient robotic learning transfer.

EDUCATION

Stanford University, Stanford, CA, USA

Ph.D. Candidate in Mechanical Engineering, Robotics 2023 – Present

Adviser: Prof. Mark Cutkosky

M.S. in Mechanical Engineering, Robotics & Control Track

Adviser: Prof. Fei-Fei Li, Prof. Jiajun Wu

Purdue University, West Lafayette, IN, USA

B.S in Mechanical Engineering 2019 – 2021

Adviser: Prof. Karthik Ramani

Shanghai Jiao Tong University, Shanghai, China

B.S in Mechanical Engineering 2017 – 2019

Tsien Hsue-Shen Honor Program

AFFILIATIONS

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Member, Stanford Center for Design Research (CDR)

Member, Biomimetics & Dexterous Manipulation Lab (BDML)

Professional Member, Association for Computing Machinery (ACM)

Professional Member, Institute of Electrical and Electronics Engineers (IEEE)

Member, Sigma Xi, The Scientific Research Honor Society

PUBLICATIONS

(*Indicate Equal Contribution)

- [1] **Li, H.***, Chung, L. Y.*, Goler, J., Zhang, R., Xie, X., Ha, H., Song, S., & Cutkosky, M. R. (2026). UMI-Underwater: Learning Underwater Manipulation without Underwater Teleoperation. In *Robotics: Science and Systems (RSS 2026)*.
- [2] Ren, T.-A., Garcia, J., Hong, S., Grinberg, J., Choi, H., Di, J., **Li, H.**, Grinberg, D., & Cutkosky, M. R. (2026). Multimodal Sensing for Robot-Assisted Sub-Tissue Feature Detection in Physiotherapy Palpation. In *Proceedings of the 2026 Design of Medical Devices Conference (DMD2026)*.

- [3] Wang, Y., **Li, H.**, Zhu, Y., Yu, K., Goldberg, K., Fei-Fei, L., Wu, J., Li, Y., & Zhang, R. IMPASTO: Integrating Model-Based Planning with Learned Dynamics Models for Robotic Oil Painting Reproduction. In *2026 IEEE International Conference on Robotics and Automation*.
- [4] Thomasson, R., Bernardini, A., **Li, H.**, Xing, C., Hajj-Ahmad, A., & Cutkosky, M. R. (2025). SLIM: A Symmetric, Low-Inertia Manipulator for Constrained, Contact-Rich Spaces. *IEEE Robotics and Automation Letters*, 10(9), 8682–8689. doi:10.1109/LRA.2025.3585712.
- [5] Lin, Y., Wei, Y. L., Liao, H., **Li, H.**, Xing, C., & Cutkosky, M. R. (2025). TypeTele: Releasing Dexterity in Teleoperation by Dexterous Manipulation Types. In *Conference on Robot Learning (CoRL 2025)*.
- [6] Xing, C.*, **Li, H.***, Wei, Y. L., Ren, T. A., Tu, T., Lin, Y., ... & Cutkosky, M. R. (2025). TacCap: A Wearable FBG-Based Tactile Sensor for Seamless Human-to-Robot Skill Transfer. *IEEE/RSJ International Conference on Intelligent Robots and Systems* (2025).
- [7] **Li, H.***, Xing, C.*, Khan, S., Zhong, M., & Cutkosky, M. R. (2024). Whisker-Inspired Tactile Sensing: A Sim2Real Approach for Precise Underwater Contact Tracking. *IEEE Robotics and Automation Letters* (2025).
- [8] Lin, M.A., **Li, H.**, Xing, C., & Cutkosky, M. (2024). Navigation and 3D Surface Reconstruction from Passive Whisker Sensing. *10.48550/arXiv.2406.06038* (2024).
- [9] Wei, Y.L., Jiang, J.J., Xing, C., Tan, X., Wu, X.M., **Li, H.**, Cutkosky, M., & Zheng, W.S. (2024). Grasp as You Say: Language-guided Dexterous Grasp Generation. In *38th Annual Conference on Neural Information Processing Systems*.
- [10] Ipsita, A.*, Duan, R.*, **Li, H.***, Chidambaram, S., Cao, Y., Liu, M., Quinn, A., & Ramani, K. (2023). The Design of a Virtual Prototyping System for Authoring Interactive Virtual Reality Environments From Real-World Scans. *ASME Journal of Computing and Information Science in Engineering*, March 2024; 24(3): 031005.
- [11] Gao, R.*, Dou, Y.*, **Li, H.***, Agarwal, T., Bohg, J., Li, Y., Fei-Fei, L., & Wu, J. (2023). The OBJECTFOLDER BENCHMARK: Multisensory Object-Centric Learning with Neural and Real Objects. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- [12] Gao, R.*, **Li, H.***, Dharan, G., Wang, Z., Li, C., Xia, F., Savarese, S., Fei-Fei, L., & Wu, J. (2023). SONICVERSE: A Multisensory Simulation Platform for Embodied Household Agents that See and Hear. In *2023 IEEE International Conference on Robotics and Automation*.
- [13] **Li, H.***, Zhang, Y.*, Zhu, J., Wang, S., Lee, M.A., Xu, H., ... & Wu, J. (2022). See, Hear, and Feel: Smart Sensory Fusion for Robotic Manipulation. In *6th Annual Conference on Robot Learning*.

- [14] Ipsita, A., **Li, H.**, Duan, R., Cao, Y., Chidambaram, S., Liu, M., & Ramani, K. (2021). VRFromX: From Scanned Reality to Interactive Virtual Experience with Human-in-the-Loop. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (pp. 1–7).

AWARDS & SCHOLARSHIPS

- HAI GenAI+Learning Grant, Stanford University 2025
- HAI Seed Research Grant, Stanford University 2024
- Howard L. Timms Scholarship, Purdue University 2020
- Dean’s List & Semester Honors, Purdue University 2020

SERVICES

Conference Reviewer

- International Conference on Computer Vision (ICCV), 2025
- Conference on Intelligent Robots and Systems (IROS), 2025, 2026
- Cognitive Science Conference (CogSci), 2025, 2026
- Conference on Computer Vision and Pattern Recognition (CVPR), 2025
- International Conference on Human-Robot Interaction (HRI), 2025
- International Conference on Robotics & Automation (ICRA), 2025
- International Conference on Learning Representations (ICLR), 2025, 2026
- Conference on Robot Learning (CoRL), 2023, 2024
- Conference on Human Factors in Computing Systems (CHI), 2024, 2025, 2026
- Winter Conference on Applications of Computer Vision (WACV), 2026
- Robotics: Science and Systems (RSS), 2025
- British Machine Vision Conference (BMVC), 2026

Journal Reviewer

- IEEE Robotics and Automation Letters (RA-L)

Seminar Organizer

- Stanford Robotics Seminar

TEACHING

- Course Assistant, Stanford University** Spring 2023
- CS231N: Deep Learning for Computer Vision
- Course Assistant, Stanford University** Fall 2022
- AA274A: Principle of Robot Autonomy

MENTORSHIP

- Stanford University** 2021 – Present
- Undergraduate Student Mentor
 - Saad Khan, Ziyang Chang
- Graduate Student Mentor
 - Tanmay Agarwal, Tianyu Tu, Clive Chung, Juhyun Jung, Yuqian Ye, Haojun Feng, Jerry Cheng, Alan Zhao, Xiaochi Xie

PRESS COVERAGE

- Columbia Broadcasting System (CBS)** 2024
- Stanford Robotics Center to develop new technology across wide range of fields.
- San Francisco Chronicle** 2024

- Stanford lab is pioneering robots to help with everything from chores to home medical care

Stanford University

2024

- Tour state-of-the-art Stanford Robotics Center

SKILLS

Real Robot Experience:

- Flexiv Rizon 4 Arm, Franka Emika Panda Arm, Turtlebot, Qysea Fifish

Design and Prototyping:

- OnShape, SOLIDWORKS, Unity, ANSYS, Pybullet, ROS

Programming:

- Python, C#, C++, Arduino, LaTeX

[CV compiled on 2026-05-08]